Environment Encoding for AGV Navigation Using Vision

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ABSTRACT

This paper presents a new "pose" (position orientation and scaling estimation) recovery method for an Automated Guided Vehicle (AGV) moving in an encoded environment using vision. The environment is structured by having large surfaces on the floor permanently encoded with the elements of a pseudo-random binary array (PRBA).

INTRODUCTION

Advanced Automated Guided Vehicles (AGVs) which can move in any direction on the floor without any path-guide to track are currently in an experimental stage, [1]. Two-dimensional (2-D) position recovery is essential for free-ranging AGV navigation and vision is the most powerful method to implement this feature. A number of machine vision methods use position referencing patterns,[2] and [3], but their applicability is restricted to a limited number of *a priori* defined locations which finally does not allow for the AGV position recovery at any point on the floor.

This paper discusses a new method to recover the AGV's absolute position and orientation anywhere on the floor using vision. The floor is permanently encoded with the terms, marked by distinct graphical symbols, of a 'pseudo-random binary array' (PRBA). As the requested image processing is limited to recognition of the binary symbols, the proposed solution is considerably faster and more cost efficient than other vision techniques employed for AGV position recovery.

PSEUDO-RANDOM ARRAY ENCODING

The position measurement paradigm, illustrated in Fig. 1, can be formally stated as follows: "Given a PRBA encoded floor, whose grid node coordinates are defined in a 2-D floor reference frame $x_uO_uy_u$, and given a perspective image showing only a part of the PRBA, determine the 2-D position and orientation of the AGV's frame $x_rO_ry_r$ in the floor's frame $x_uO_uy_u$."

The floor is permanently encoded with the elements of a PRBA. Such an *n1*-by-*n2* array can be obtained as shown in Fig. 2, by folding a (2^n-1) -term pseudo-random binary

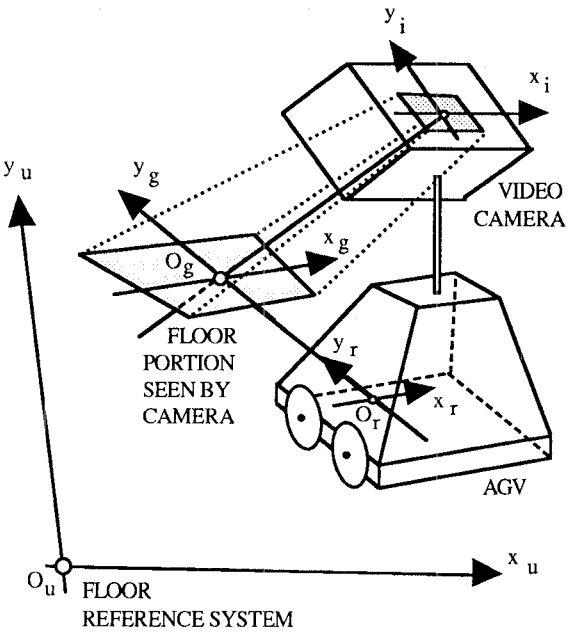


Fig. 1. The free-ranging AGV position recovery paradigm

sequence { S(p) | p= 0,1, ..., 2^n-2 } generated by an n-bit shift register having a modulo-two feedback. The following relations hold for this PRBA, [4]:

$$2^n-1 = 2^{k1 \cdot k2}-1 \quad (1)$$

$$n1 = 2^{k1}-1 \quad (2)$$

$$n2 = (2^n-1)/n1 \quad (3)$$

where *n1* and *n2* must be relatively prime.

The 2-D coordinate recovery is based on the PRBA window property. According to this property, any *k1*-by-*k2* nonzero binary pattern seen through a *k1*-by-*k2* window

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sliding over the array is unique and may fully identify the window's absolute coordinates (i, j) within the PRBA. The "pseudo-random / natural" code conversion is implemented as a memory stored table.

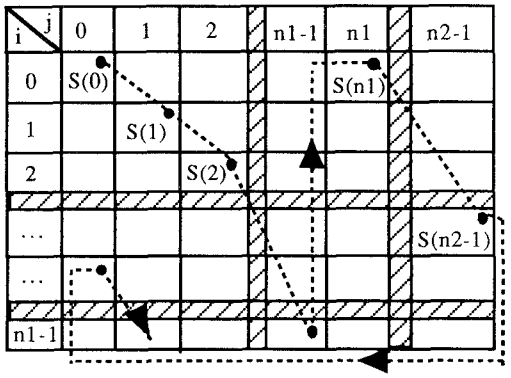


Fig. 2. A (2^n-1)-bit PRBS $\{S(p) \mid p= 0,1, ..., 2^n-2\}$ is folded to produce an $n1$-by-$n2$ PRBA

For instance, as shown in Fig. 3, a 63-bit sequence (generated by an n=6 bit shift register) will produce a PRBA with n1=7 lines (since k1=3, and k2=2) and n2=63/7=9 columns. The pseudo-random binary pattern seen through a 3-by-2 window sliding over this array is unique and may be used to identify the window's absolute coordinates (i, j).

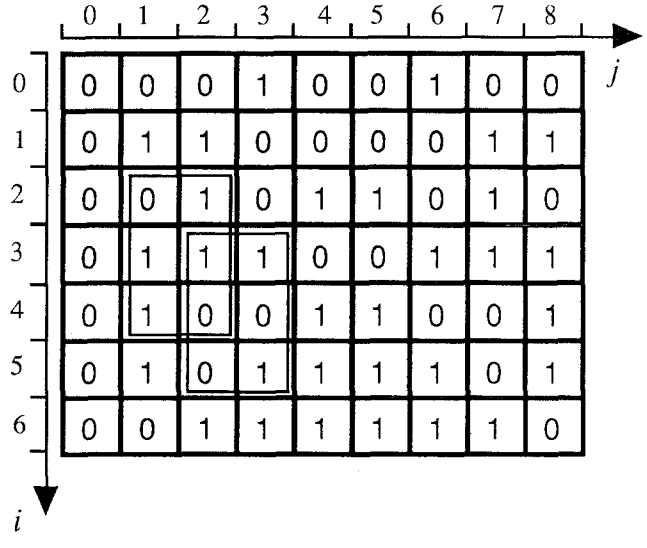


Fig. 3. Example of the window property in a PRBA

Visual Recognition of the Binary Symbols

Two distinct graphical symbols, shown in Fig. 4, have been used to mark the binary values "1" and "0" within the PRBA. The particular shapes of these binary symbols are selected in such a way as to meet the following requirements: (i) there is enough information at the symbol level to provide an immediate indication of the array's orientation and about where neighboring symbols can be found; (ii) the symbol recognition procedure is invariant to position, orientation, scaling, and perspective transformations; (iii) the symbols should have sufficient uniqueness so that other objects in the scene won't be mistaken for encoding symbols.

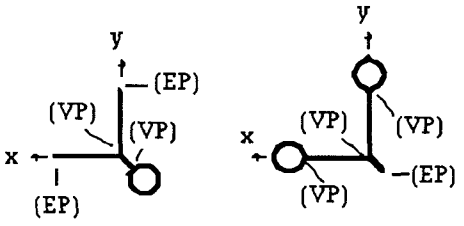


Fig. 4. The two graphical symbols used to mark "0" and "1" within the PRBA. The EP notation identifies an end point and VP identifies a vertex point in the skeleton of these symbols.

The binary symbol recognition is done by finding the skeleton of the binary image. The symbols are recognized by their numbers of end-points and vertices. The symbol representing "0" has 2 end-points and 2 vertex-points and the symbol for "1" has 1 end-point and 3 vertex-points.

ABSOLUTE POSITION AND ORIENTATION RECOVERY

Perspective Image Geometry

Given the 2-D coordinates $\{x(A^i), y(A^i)\}$ of a point A in the image frame $x_iO_iy_i$, and knowing the camera's tilt angle α and its lens distance O_iO_g (along the optical axis) to the floor, the problem is to find the coordinates $\{x(C^g), y(C^g)\}$ of the point C (the point A's projection on the floor) in the reference frame $x_gO_gy_g$ (which is the floor projection of the image reference frame).

Considering an intermediate reference system $x_gO_gy_p$ which defines a plane parallel to the image plane, as shown in Fig. 5, we find the triangles FTO_g and FSO_i to be similar, which gives:

$$O_gT/O_iS = FO_g/FO_i = m \qquad (4)$$

where "m" is a constant magnification ratio. Using coordinate notation, we obtain from (4):

$$x(C^g) = m \cdot x(A^i) \qquad (5)$$

We also find the triangles FRO_g and FPO_i to be similar, which gives:

$$O_gR/O_iP = FO_g /FO_i = m \qquad (6)$$

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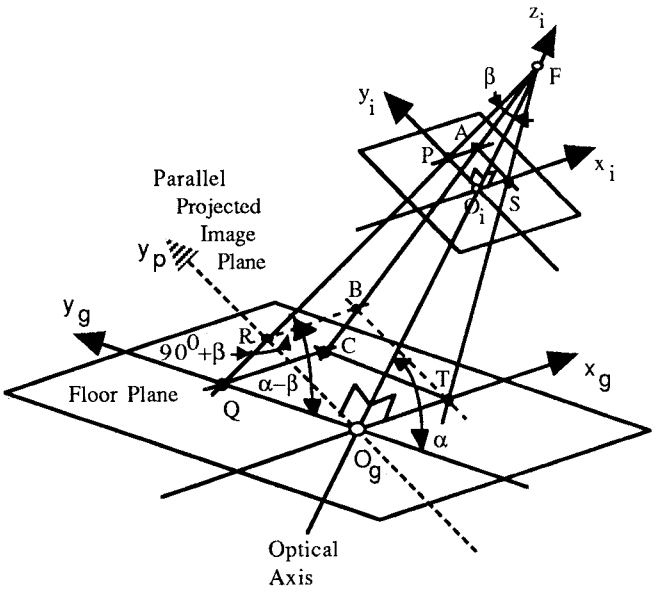


Fig. 5. Perspective distortions due to camera tilt (angle α)

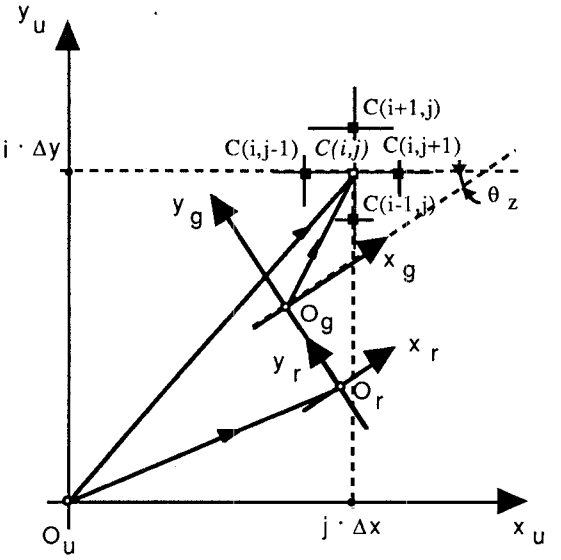


Fig. 6. The position recovery reduced to the floor plane.

Applying the law of sines to the triangle RQO_g we obtain:

$$O_gQ = m \cdot O_iP / (\sin\alpha - \cos\alpha \cdot \tan\beta) \quad (7)$$

From the triangle FO_iP:

$$\tan\beta = PO_i/FO_i = y(A^i)/f \quad (8)$$

where "f" is the constant focal length of the camera lens.

Replacing $\tan\beta$ given by (8) in (7) and using coordinate notation we finally obtain:

$$y(C^g) = y(A^i) \cdot m \cdot f / [f \cdot \sin\alpha - \cos\alpha \cdot y(A^i)] \quad (9)$$

2-D Position and Orientation Recovery

As illustrated in Fig. 6, the AGV's orientation is defined by the angle θ_z between the x-axes of the robot's, and respectively floor's, reference frames. Because of the parallelism which exists (by construction) between the projected image reference frame and the robot's frame on one side, and between the PRBA grid lines and the axes of the the floor's reference frame on the other side, this orientation angle θ_z is identical with the angle between any grid line in the reconstructed PRBA and the O_gx_g axis. It is then possible to write from Fig. 6:

$$\tan\theta_z = [y(C^g(i,j)) - y(C^g(i,j+1))] / [x(C^g(i,j)) - x(C^g(i,j+1))]; \quad (10)$$

Using (5) and (9) which relate the 2-D coordinates of the floor marked grid nodes $C^g(...,...)$ to those of their images $A^i(...,...)$, the relation (10) finally becomes:

$$\tan\theta_z = \frac{y(A^i(i,j)) - y(A^i(i,j+1))}{x(A^i(i,j)) - x(A^i(i,j+1))} \cdot \frac{f^2 \cdot \sin\alpha}{[f \cdot \sin\alpha - \cos\alpha \cdot y(A^i(i,j))] \cdot [f \cdot \sin\alpha - \cos\alpha \cdot y(A^i(i,j+1))]}; \quad (11)$$

which allows calculation of the AGV's orientation from the 2-D coordinates of two adjacent grid nodes, $A^i(i,j)$ and $A^i(i,j+1)$, identified in the PRBA image acquired by the onboard video camera.

The following vectorial equation can be written for the relationship which exists between the frames of the AGV, projected image, and floor:

$$\overline{O_uC(i,j)} = \overline{O_uO_r} + \overline{O_rO_g} + \overline{O_gC(i,j)} \quad (12)$$

The following two relations result for the 2-D coordinates of the AGV's frame origin O_r in the floor reference frame:

$$x(O_r^u) = j \cdot \Delta x + d \cdot \sin\theta_z - m \cdot x(A^i(i,j)) \cdot \cos\theta_z + m \cdot f \cdot y(A^i(i,j)) \cdot \sin\theta_z / [f \cdot \sin\alpha - \cos\alpha \cdot y(A^i(i,j))]; \quad (13)$$

$$y(O_r^u) = i \cdot \Delta y - d \cdot \cos\theta_z - m \cdot x(A^i(i,j)) \cdot \sin\theta_z - m \cdot f \cdot y(A^i(i,j)) \cdot \cos\theta_z / [f \cdot \sin\alpha - \cos\alpha \cdot y(A^i(i,j))]; \quad (14)$$

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where $d=O_rO_g$ is the fixed distance between the AGV's frame origin and the projected image's frame origin, and Δx and Δy are the quantization steps on the "x" and "y" directions respectively.

The last two relationships allow calculation of the AGV's position from the 2-D coordinates of a grid node $A^i(i,j)$ identified in the PRBA image acquired by the onboard video camera. The other parameters to be used are "m" which is the magnification ratio, "f" which is the focal length, α which is the camera's tilt angle, the AGV's orientation angle θ_Z previously recovered using (11).

EXPERIMENTAL RESULTS

The experimental AGV, shown in Fig. 7, was developed to validate the discussed vision method for 2-D absolute position recovery. The floor is permanently marked (with a 55 mm grid-step) with the terms of a 63-by-65 PRBA (with k1=6, k2=2, n1=63, n2=65) which requires a 6-by-2 window for absolute position recovery.

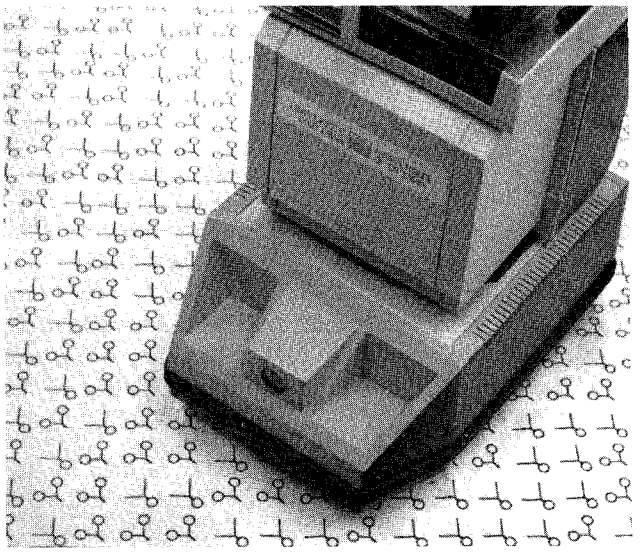


Fig. 7. The experimental vision guided AGV on the PRBA encoded floor

Experimental results shown in Fig.8 illustrate the performance of this technique. In this case, the recovered grid node coordinates in the floor reference system $x_uO_uy_u$ are: i= 2 and j= 60, and the AGV's orientation angle is $\theta_Z = 5.71^0$.

As expected, the production of the skeleton is the most time-consuming part. Extreme perspective distortion can result in wrong neighbor distance estimation. The method is extremely robust because it never produced a grid with false symbols. Symbols may not be recognized, or a "1" symbol may be recognized as a "0" symbol due to a break of the skeleton, and sometimes noise may be recognized as a symbol. However,

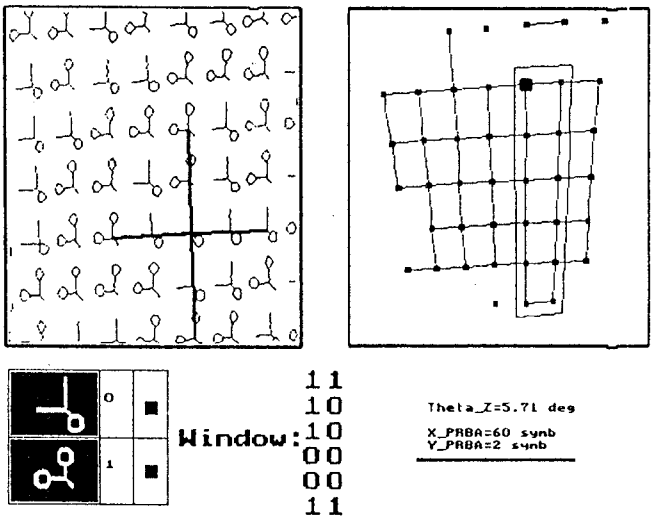


Fig. 8. Floor image acquired by the onboard camera, and the recovered positional parameters

during testing, these falsely recognized symbols never had the right size and orientation to fit to their neighbors as required for acceptance and integration into the PRBA grid.

CONCLUSION

The ability to instantaneously recover 2-D absolute position allows a free-ranging AGV to move on any direction on the encoded floor . The PRBA-based encoding technique requires only one bit of code per quantization interval, independent of the desired resolution. The onboard camera also helps the AGV's navigation by providing a larger and deeper perspective of the environment .

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